

Daris Chen

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EDUCATION

University of California San Diego

Bachelor of Science (B.S.) in Computer Engineering

March 2026
San Diego, CA

WORK EXPERIENCE

Lead Full-Stack Engineer

Jan 2026 – Jun 2026

Product Manager Accelerator

Remote

- Constructed a RAG-based patent search and drafting assistant deployed end-to-end on Vercel, AWS EC2, and Supabase using Next.js, exposing FastAPI REST endpoints with Auth0 authentication to simplify intellectual property research for early-stage founders.
- Programmed a hybrid semantic search engine over 12,000+ USPTO patents combining OpenAI embeddings with PostgreSQL pgvector and HNSW indexing, achieving sub-100ms query latency for real-time prior-art discovery.
- Led a 14-member cross-functional team as lead architect and primary contributor, running Agile workflows in ClickUp and assigning work by strength and task atomicity to sustain a 100% no-blocker rate on daily standups across a six-week MVP.

Lead Full-Stack Engineer

Aug 2025 – Mar 2026

Groundwork Books

San Diego, CA

- Led an in-place migration of a production e-commerce storefront to Next.js and Vercel under live customer traffic, integrating the Square API so storefront stock reflects point-of-sale inventory without manual updates and modernizing the tech stack.
- Engineered a semantic search layer using Pinecone with integrated text embeddings, enabling topic and concept-based discovery across 4,000+ SKUs without requiring exact title matches.
- Architected a two-tier cache: a look-aside Redis reducing Square API latency by 90% (sub-200ms p99), and a client-side request coalescing hook batching 50+ concurrent inventory lookups into a 75 ms window API call with 2-minute TTL.

AI Researcher

Feb 2025 – Mar 2026

UCSD Engineers for Exploration (E4E) FishSense

San Diego, CA

- Rebuilt a published RRPN laser-detection network (ResNet-50 backbone) to a working training and inference state in Detectron2, pinning a brittle CUDA and PyTorch dependency matrix into a reproducible setup for the research group.
- Devised a dual-laser photogrammetry algorithm to extract biological metrics (fish length/mass) from 2D video feeds, achieving sub-5% error rates in pixel-to-centimeter conversion.

PROJECTS

Phased Array Microwave Sensor | *ESP32-S3-DevkitC-1, Raspberry Pi 4, Analog Design*

Feb 2026 – Apr 2026

- Constructed an analog front-end for a 4x1 HB100 microwave transceiver array, cascading dual-stage MCP6002 op-amps for 1092x gain on microvolt IF signals with 1.65V virtual-ground biasing to read bipolar signals on a unipolar ADC.
- Programmed a processing pipeline combining 4th-order Butterworth filters and FFTs with MUSIC and an Extended Kalman Filter for direction estimation and 2D spatial tracking.
- Coded an amplitude monopulse zone classifier deriving spatial bias and target velocity, substituting for phase-coherent direction finding to work around inherent local-oscillator drift in the hardware.

TriCore9 | *SystemVerilog, Python, ModelSim, Quartus Prime*

Sep 2025 – Dec 2025

- Designed a custom 8-bit RISC processor in SystemVerilog with a 9-bit instruction format, 4-register ISA, 8-function ALU, and 256-byte data memory, resolving fetch, decode, ALU, and data-memory stages within a single cycle.
- Wrote a Python assembler encoding R-type, load-immediate, and branch instructions to machine code, validating the ISA through a Hamming-code test program and two arithmetic programs in ModelSim.

16 bit Carry Look Ahead Adder | *Verilog, Digital (hneemann), Digital Logic Design (Digital)*

Sep 2024 – Dec 2024

- Constructed a hierarchical 16-bit adder in Digital from cascaded 4-bit propagate-generate blocks, reducing critical path delay from $O(N)$ to $O(\log N)$ against a ripple-carry baseline.
- Implemented a 2421 BCD adder variant, validating logical correctness through exhaustive corner-case simulation.

TECHNICAL SKILLS

Programming Languages: Verilog/SystemVerilog, C/C++, Python, ARM Assembly, SQL, JavaScript/TypeScript

Hardware & Embedded: ESP32-S3, Raspberry Pi, HB100, RF engineering, analog front-end design, op-amp cascades, virtual-ground biasing, ADC interfacing, DMA, interrupt handling, FPGA synthesis, timing analysis

DSP & Signals: FFT, Butterworth filtering, MUSIC angle-of-arrival, Extended Kalman Filter, Doppler processing, amplitude monopulse, digital filter design

Tools & Technologies: Intel FPGA ModelSim, Quartus Prime, Git, GitHub Actions, Linux, OpenCL, CUDA, PyTorch, NumPy, Jupyter, Docker

Databases: PostgreSQL, MongoDB, Firebase, Supabase